6. Represent the following sets in descriptive form.

(i) P ={ January, June, July}

(ii) Q = {7,11,13,17,19,23,29}

(iii) R = $\{x : x \in \mathbb{N}, x < 5\}$

(iv) S = {x : x is a consonant in English alphabets}

1.4 Types of Sets

There is a very special set of great interest: the empty collection ! Why should one care about the empty collection? Consider the set of solutions to the equation $x^2+1 = 0$. It has no elements at all in the set of Real Numbers. Also consider all rectangles with one angle greater than 90 degrees. There is no such rectangle and hence this describes an empty set.

So, the empty set is important, interesting and deserves a special symbol too.

1.4.1 Empty Set or Null Set

A set consisting of no element is called the *empty set* or *null set* or *void set*.

It is denoted by ∅ or { }.

Thinking Corner

Are the sets {0} and {∅} empty sets?

For example,

(i) A={x : x is an odd integer and divisible by 2}

$\therefore A$={ } or ∅

(ii) The set of all integers between 1 and 2.

1.4.2.Singleton Set

A set which has only one element is called a *singleton set*.

For example,

(i) $A = \{x : 3 < x < 5, x \in \mathbb{N}\}$ (ii) The set of all even prime numbers.

1.4.3 Finite Set

A set with finite number of elements is called a *finite set*.

Note

An empty set has no elements, so ∅ is a finite set.

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